

NumPy.ipynb X

C:\> Users > Sai > Documents > Python Jupyter > NumPy.ipynb > import numpy as np

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[1] `import numpy as np`

[2] `print(np.__version__)`

--- 2.4.1

[3] `## creating arrays`

[4] `list1 = [1,2,3]`
`print(list1)`

--- [1, 2, 3]

[5] `arr_list = np.array(list1)`
`print(arr_list)`

--- [1 2 3]

3 2

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```
arr_list = np.array(list1)
print(arr_list)
```

[5]

```
[1 2 3]
```

```
print(type(arr_list))
```

[6]

```
<class 'numpy.ndarray'>
```

```
list1 = [1,2,3]
list2 = [4,5,6]
arr_2d = np.array([list1, list2])
print(arr_2d)
```

[7]

```
[[1 2 3]
 [4 5 6]]
```

```
list1 = [1,2,3]
list2 = [4,5,6]
list3 = [7,8,9]
```

[8]

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```
list1 = [1,2,3]
list2 = [4,5,6]
list3 = [7,8,9]
arr_3d = np.array([[list1, list2, list3]])
print(arr_3d)
```

[8]

```
[[[1 2 3]
    [4 5 6]
    [7 8 9]]]
```

```
print(arr_3d.ndim)
```

[11]

3

2 NumPy array attributes

[12]

```
arr1 = np.array([[1,2,3], [4,5,6]])
print(arr1)
```

[13]

```
[[1 2 3]
 [4 5 6]]
```

X 3 2

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```
arr1 = np.array([[1,2,3] , [4,5,6]])  
print(arr1)
```

[13]

```
...  
[[1 2 3]  
 [4 5 6]]
```

```
print("shape" , arr1.shape)  
print("size" , arr1.size)  
print("dtype", arr1.dtype)  
print("n_dim", arr1.ndim)
```

[14]

```
...  
shape (2, 3)  
size 6  
dtype int64  
n_dim 2
```

3 Array Initialization Methods

[15]

zero array

[16]

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3 Array Initialization Methods

[15]

zero array

```
zero_arr = np.zeros((2,3))  
print(zero_arr)
```

[16]

```
...  
[[0. 0. 0.]  
 [0. 0. 0.]]
```

one array

```
one_arr = np.ones((2,3))  
print(one_arr)
```

[18]

```
...  
[[1. 1. 1.]  
 [1. 1. 1.]]
```

Full Array

```
full_arr = np.full((2,3), 7)
```

[19]

3 2

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one array

```
one_arr = np.ones((2,3))  
print(one_arr)
```

[18]

```
...  
[[1. 1. 1.]  
 [1. 1. 1.]]
```

Full Array

```
full_arr = np.full((2,3), 7)  
print(full_arr)
```

[19]

```
...  
[[7 7 7]  
 [7 7 7]]
```

#Identity methods

```
id_arr = np.eye(3)  
print(id_arr)
```

[21]

```
...  
[[1. 0. 0.]  
 [0. 1. 0.]  
 [0. 0. 1.]]
```

< 3 2

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Empty Array

```
print(np.empty(2))
```

[22]

```
... [1.00132483e-307 1.61898049e-317]
```

#evenly spaced array

```
print(np.arange(1,3,1))
```

[24]

```
... [1 2]
```

#specific number of values between a range

```
print(np.linspace(1,10,4))
```

[26]

```
... [ 1.  4.  7. 10.]
```

#random values array - float

[28]

x 3 2

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#random values array - float

```
r_array = np.random.rand(3,2)
print(r_array)
```

[28]

...

```
[[0.29462416 0.54003334]
 [0.01947281 0.42042399]
 [0.664111   0.52488949]]
```

#random array - int

```
r_are = np.random.randint(1,10,(3,2))
print(r_are)
```

[30]

...

```
[[4 6]
 [8 7]
 [4 4]]
```

#4 Array Indexing and slicing

[31]

```
a = np.array([1,2,3,4,5])
```

[32]

3 2

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#4 Array Indexing and slicing

[31]

```
a = np.array([1,2,3,4,5])  
print(a)
```

[32]

```
... [1 2 3 4 5]
```

```
print(a[0])
```

[33]

```
... 1
```

```
print(a[-1])
```

[34]

```
... 5
```

```
print(a[2:3])  
print(a[2:4:2])  
print(a[-3:])
```

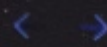
[36]

3 2

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[36]

```
print(a[2:3])  
print(a[2:4:2])  
print(a[-3:])
```

...

```
[3]  
[3]  
[3 4 5]
```

[38]

```
a2 = np.array([[1,2,3] , [4,5,6]])  
print(a2)
```

```
[[1 2 3]  
 [4 5 6]]
```

[41]

```
print(a2[0])  
print(a2[1])  
print(a2[0][1])
```

```
[1 2 3]  
[4 5 6]  
2
```



3 2



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```
print(a2[1:])  
print(a2[:])  
print(a2[1:2])
```

[45]

```
...  
[[4 5 6]]  
[[1 2 3]]  
[4 5 6]  
[[4 5 6]]
```

```
#print(a3[:, 1:2])  
#print(a3[1:2, 2:2])
```

[1]

#5 . Array Reshaping and Flattening

[2]

Generate

```
array1 = np.array([1,2,3,4,5,6])  
print(array1)
```

```
reshaped = array1.reshape((5,1))
```

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```
#print(a3[1:2 , 2:2])
```

[1]

#5 . Array Reshaping and Flattening

[2]

```
array1 = np.array([1,2,3,4,5,6])
print(array1)
```

```
reshaped = array1.reshape((5,1))
print(reshaped)
```

```
print(reshaped.flatten())
```

6. Array Stacking and Splitting

[10]

```
a = np.array([1,2,3])
```

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```
print(reshaped.flatten())
```

6. Array Stacking and Splitting

[10]

```
a = np.array([1,2,3])
```

```
b = np.array([4,5,6])
```

```
print(np.vstack((a,b))) #vertical staking
```

```
print(np.hstack((a,b))) #horizontal stacking
```

```
c = np.array([[1,2,3], [4,5,6]])
```

```
print(np.vsplit(c,3))
```

```
print(np.hsplit(c,3))
```

#7. Mathematical Operations on Array

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#7. Mathematical Operations on Array

```
a = np.array([1,2,3,-2])  
print(a)
```

```
print(a+1)  
print(a-1)  
print(a*2)  
print(a/2)
```

```
b = np.array([1,4,9])  
print(np.square(b))  
print(np.sqrt(b))
```

#homework : sin , cos , tan , sec

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#8. Mathematical operation on multiple arrays

```
a = np.array([1,2,3])  
b = np.array([4,5,6])  
  
print(np.add(a,b))  
print(np.subtract(a,b))  
print(np.multiply(a,b))  
print(np.divide(a,b))
```

```
print(np.dot(a,b)) #dot product  
print(a.T) #transpose
```

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#9 .statistical Function

```
a = np.array([[1,2,3],[4,5,6]])
```


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#9 .statistical Function

```
a = np.array([[1,2,3],[4,5,6]])  
print(a)
```

```
print(np.sum(a))  
print(np.mean(a))  
print(np.median(a))  
print(np.std(a))  
print(np.min(a))  
print(np.max(a))
```

10. Array Comparison

```
a = np.array([1,2,3])  
b = np.array([4,5,6])  
print(a == b) #comparing each array
```

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10. Array Comparison

```
a = np.array([1,2,3])
b = np.array([4,5,6])

print(a == b) #comparing each array
print(np.array_equal(a,b)) #comparing all array
```

11 Braodcasting

```
a = np.array([1,2,3])
b = np.array([4])

print(a+b)
```

12. Handling with nan & inf

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12. Handling with nan & inf

```
data1 = np.array([1,2, np.nan , ,4 , np.inf])  
print(data1)  
  
print(np.isnan(data1))  
print(np.nan_to_num(data1))
```

Save and Load array

```
import numpy as np  
  
arr = np.array([1,2,3])  
print(arr)  
npsave('my_array.npy', arr)
```

[2]

[1 2 3]

NameError

Traceback (most recent call last)

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```
import numpy as np

arr = np.array([1,2,3])
print(arr)
np.save('my_array.npy', arr)
```

[2]

```
[1 2 3]
```

NameError

Cell In[2], line 5

Traceback (most recent call last)

```
3 arr = np.array([1,2,3])
```

```
4 print(arr)
```

```
----> 5 np.save('my_array.npy', arr)
```

NameError: name 'np.save' is not defined

Generate

+ Code

```
loaded_arr = np.load('my_array.npy')
print(loaded_arr)
```

3 2

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